

SY BCA DIV-C
SUBJECT: 303 Database handling using Python (SQLite)
LAB Program List (GUIDE-1)
[2026-2027]

WEEK-1

1. Create a database <database_name> and create following table in to that. Database_name: 'STUD'.

Table-Student (rno,sname,city,age)

- 1) Create above tables by providing appropriate constraints.
- 2) In table contains minimum 10 records of student.
- 3) Display rno,name of student,city from table.
- 4) Display rno,name of student,city where city is valsad.
- 5) Display names of student,city where student name starts with 'A'.
- 6) Change the city name from valsad to vapi whose rno=5.
- 7) Delete the records of student whose rno=4.
- 8) Insert new student record for rno=4

2. Create a database STUDENT.db and create following table in to that.

Table-Student (S_no,S_name, age, City, Percentage, Class_no)

- 1) Create above table by providing appropriate constraints.
- 2) In tables contains minimum 10 records.
- 3) Display name of students coming from vapi or valsad in to student table.
- 4) Display name of students who belong to class 'c2' in to student table.
- 5) Display names of student who secured more than 70 percentage in to student table.
- 6) Display names of student who secured percentage between 70 and 60 in to student table.
- 7) Display names of student whose name start with 'A' or 'P'.
- 8) Display names of student whose second character in name is 'a' or 'i' in to student table.
- 9) Display names of student,city and percentage in ascending order percentage wise in to student table.
- 10) Display names of student,city and percentage in desending order student_name wise in to student table.

3. Create a database SALES.db and create following table in to that.

Table: Sales (S_no,S_name,S_month,City,Amount)

- 1) Create above tables by providing appropriate constraints.
- 2) In tables contains minimum 10 records.
- 3) Display total sales.in to sales table.
- 4) Display total sales occurred in month 'may' in to sales table.
- 5) Display total sales of 'Ramesh' in to sales table.
- 6) Display total sales of 'Ramesh' in June and May month in to sales table.
- 7) Display total sales of 'Ramesh' and 'Rakesh' in to sales table.
- 8) Display total sale month wise in to sales table.
- 9) Display S_name and total sales where total sale value is more than 50,000 in to sales table.

4. Create a database fybca_Student.db and Create following table and add records in the table. Student (StudentID INTEGER, Primary Key,FirstName TEXT, Not Null,LastName TEXT, Not Null,Gender TEXT, Check constraint (Must be 'Male' or 'Female'),DateOfBirth TEXT, Not Null,Grade INTEGER, Check constraint (Must be between 1 and 12),Email TEXT, Unique)

- 1) Create above table by providing appropriate constraints.
- 2) In tables contains minimum 10 records.
- 3) Write the SQL command to select all columns and all records from the 'Students' table.
- 4) Write the SQL command to select only 'FirstName', 'LastName', and 'Grade' from the 'Students' table.

- 5) Write the SQL command to select all students who are in Grade 10.
- 6) Write the SQL command to select all female students who are in Grade 9 or above.
- 7) Write the SQL command to select all students and order them by `LastName` in ascending order.
- 8) Write the SQL command to count the number of students in each grade and display the result.
- 9) Write the SQL command to select all students who were born after January 1, 2006.
- 10) Write the SQL command to select all students whose email ends with `@example.com`.
- 11) Write the SQL command to update the grade of `Jayesh` to 11.
- 12) Write the SQL command to delete the record of the student with the email `jayesh11@example.com`.
- 13) Write the SQL command to find the youngest student in the `Students` table.
- 14) Write the SQL command to find the oldest student in the `Students` table.
- 15) Write the SQL command to count the number of students grouped by gender.

5. Create a database SALES2.db and Create following table and add records in the table.

Table: Sales (S_no, S_name, S_month, City, Amount)

- 1) Create a database <database name> and create following table in to that.
- 2) Create above tables by providing appropriate constraints.
- 3) In tables contains minimum 10 records.
- 4) Display total sales.in to sales table.
- 5) Display total sales occurred in month 'may' in to sales table.
- 6) Display total sales of 'Ramesh' in to sales table.
- 7) Display total sales of 'Ramesh' in June and May month in to sales table.
- 8) Display total sales of 'Ramesh' and 'Rakesh' in to sales table.
- 9) Display total sale month wise in to sales table.
- 10) Display S_name and total sales where total sale value is more than 50,000 in to sales table.

WEEK-2

6. Create a database class.db and Create following table and add records in the table.

Table: Class (C_no, Branch (BBA, BCA, B.com, BA), Year (FY, SY, TY), Total_student)

- 1) Create a database <database name> and create following table in to that.
- 2) Create above table by providing appropriate constraints.
- 3) In tables contains minimum 10 records.
- 4) Display name of branch and total strength for particular branch in to class table.
- 5) Display year wise total strength of students in to class table.
- 6) Display year and strength for BCA in ascending order strength wise in to class table.
- 7) Display name of branch and year for those years where strength is more than '110' in to class table.
- 8) Display name of branch and total strength where total strength is more than '400' in to class table.
- 9) Display total strength for all the branches except 'BBA' in to class table.
- 10) Display name of branch, year and total strength for those branches where first letter in name starts with 'B' in to class table.
- 11) Display name of branch, year and total strength for those branches where second letter start with 'C' in to class table.

7. Create a database result.db and Create following TABLE: Result (s_id INTEGER autoincrement primary key,s_name TEXT check (length () <25),sub1 integer check >=0 and <=100,sub2 integer check >=0 and <=100,sub3 integer check >=0 and <=100,sub4 integer check >=0 and <=100,Sub5 integer check >=0 and <=100)

- 1 INSERT 10 records in Result Table
- 2 Display all students result detail with total and percentage after adding new columns total and percentage.
- 3 Display Students name who passed in all subjects.
- 4 Display students name who failed in at least one subject.

- 5 Display student details in descending order of total.
- 6 Display student name who scores highest total.
- 7 Display student name who scores second highest total.
- 8 Display student name who scores second lowest total.
- 9 Display student detail scores highest marks in sub1.
- 10 Display student detail scores lowest marks in sub5.
- 11 Display student detail scores 50 to 70 marks in sub4.
- 12 Display student detail where student names not start with p or r.
- 13 Display student detail where student names start with p or r and second last character is s.
- 14 Write a case statement to display student name, total, percentage and Grade using Case Statement.
Per>=80 'A+'
Per>=70 and <80 'A'
Per>=60 and <70 'B'
Per>=50 and <60 'C'
Per>=35 'D'
Else 'Fail'.

8. Create database MYDB.db and perform following DDL and DML tasks with apply appropriate constraint.

- 1) Create Table STUD_MARKS contain following attributes (Sno, Sname, Sub1, Sub2, Sub3, Total, Percentage)
- 2) Insert Records in STUD_MARKS
- 3) To view Records from Table STUD_MARKS
- 4) Update total and percentage attributes.
- 5) To Delete Specific records STUD_MARKS
- 6) To remove the Table STUD_MARKS